The Integrated Toroidal–Syntropic Model: Core Architecture, Conditional Weak-Field Limit, and Open Research Gates

Brendon Boyd
Independent Researcher, Western Australia

Working draft (after 12.0-alpha.5) — July 29, 2026

Research-stage status. This document reconstructs a coherent ITSM field hierarchy and records both successful and failed research gates. It is not a completed alternative to ΛCDM and does not claim that the open matching, screening, lensing, disk, perturbation, or driven-anisotropy problems have been solved.

Abstract

The Integrated Toroidal–Syntropic Model (ITSM) is reconstructed as a layered research programme rather than a single scalar interpolation formula. The minimal canonical condensate has a stable finite-density branch, but explicit amplitude integration proves that its leading static-gradient response is quadratic, so UVIR-001 rejects it as the origin of the required spatial $Y^{3/2}$ operator. UVIR-002 provisionally separates the complex condensate's density, circulation and topology roles from a preferred-frame force scalar. UVIR-003 Stage A supplies a complete two-derivative frame basis and a regulated force truncation; its flat decoupling checks pass, while a verified Einstein–aether substitution supplies normalized frame-sector mode speeds. The zero-gradient force block also factorizes at quadratic order into one nonnegative $z = 2$ scalar, but its normalization remains unmatched. A scalar ADM readiness audit finds that the declared Minkowski finite-density background is off shell: its nonzero condensate enthalpy cannot be cancelled by a constant vacuum-energy subtraction. A background-completion screen also rejects a healthy two-derivative $P(X)$ scalar and the ghost-condensate point as exact Minkowski support, while rigid support is decoupling-only. The least-assumptive route is therefore a self-consistent evolving flat-FRW background. Its minisuperspace equations and a representative dimensionless on-shell branch are now verified, removing the background blocker. A subsequent aether-unitary principal-symbol reduction eliminates the scalar lapse and shift constraints, independently recovers the exact aether spin-0 speed and gives a positive representative high-wavenumber block above a new q_{ADM} validity scale. The complete time-dependent quadratic finite-wavenumber reduction also has positive kinetic inertia throughout its representative nonzero-wavenumber scan. Although its exact determinant scales as q^2_{phys}, the exact null direction at the homogeneous endpoint is $(H, \dot{\rho}, \mu)$, tangent to a homogeneous time translation of the FRW background. Two gauge-invariant matter combinations retain a positive regular $q = 0$ kinetic block across the representative trajectory. The apparent rank loss is therefore a gauge endpoint, not a physical homogeneous scalar. A bounded khronon expansion now derives the complete three-dimensional flat-decoupling cubic and quartic aether bases. The quartic action has 96 expanded monomials and an independent longitudinal cross-check. Its exact elastic contact term is finite and elastic t/u cubic exchange vanishes, but the centre-of-mass s channel is the non-invertible homogeneous khronon gauge orbit. Cubic reduction needs first-order constraints only, whereas quartic reduction requires the second-order constraint Schur complement. The physical interaction scale therefore remains open pending the complete gauge-regular constrained cosmological $2 \to 2$ amplitude and physical eigenmode projection. The exact nonlinear $g + U + \Phi$+alignment ADM parent block now reproduces the FRW and finite-wavenumber quadratic constraint data. Track A adopts the rest-space Laplacian, retains exact $Y^{3/2}$ and assigns its perturbative force test to a declared local nonzero-gradient background. The homogeneous force action is verified through direct quartic order. Its origin-linear source is a verified component, while the complete dressed source $S_2 = \partial_z L_3[x, z_1]$, generic $L_4[x, z_1]$ contact and corrected quartic Schur functional are assembled on the homogeneous zero-gradient Track-A branch. A regular physical basis separates the preferred-time gradient mode from two gauge-invariant matter scalars without restoring the homogeneous gauge orbit. The complete analytic cubic and reduced quartic functionals are now polarized into factorized finite-wavenumber physical-basis kernels with exact external lapse and shear resolvers. The quartic result derives the complete two-leg constraint source and all three finite-channel Schur pairings. At exact zero internal momentum, separate projectors remove the nonexistent homogeneous scalar shift and the preferred-time Ξ gauge orbit while retaining the lapse constraint and the (Q_ρ, Q_χ, Π) physical subspace. This algebraic rule is not a naive continuation of the finite-q inverse. The exact $|\nabla\pi|^3$ functional still has no ordinary Taylor kernel at zero gradient. The physical inverse quadratic kernels are constructed, with positive kinetic inertia and a clean real-pole, positive-residue regime at high momentum. The fixed-comoving follow-on yields converged time-dependent transfer matrices and invalidates frozen-pole exponentiation in the nonadiabatic domain. The deepest sampled infrared mode retains a 1.38×10^{27} normalized phase-space gain. A five-case mode-resolved follow-on finds an Ξ-seeded maximizing input in every tested case, but each trajectory enters an off-axis complex quartet whose real invariant space is rank four. No unique continuous real rank-two split between gauge-continuation and retained-matter pole pairs exists there. The subsequent source-to-observable audit restricts generalized impulses and readouts to (Q_ρ, Q_χ) and removes direct Ξ and homogeneous time-translation source support. All five cases retain amplified through-quartet matter response from 2.68×10^{17} to 9.76×10^{19}. This resolves the direct gauge-source attribution question on the tested dimensionless neighborhood, but is not an all-background instability theorem, amplitude, or cutoff. The Track-A force mode remains factorized from this coupled quadratic block. The propagating exchange-plus-reduced-contact $2 \to 2$ amplitude remains open. The force dispersion also has unresolved metric-light-cone crossings and the available longitudinal IR NDA scale is not a physical EFT cutoff. The conditional cubic force action still yields a square-root acceleration in a quasi-static high-symmetry limit, and the validated CBR-001 calculations establish topology-dependent Casimir pressure while finding $H_t/H_p = 13/12$ only as a tunable transient in the tested free-field model. Strong phenomenology therefore remains downstream of the explicit UVIR, matter, screening, lensing, disk and cosmological gates.

Contents

1 Scope, claim status, and reconstruction rule

The preceding ITSM manuscript combined an active-superfluid ontology, a Born–Infeld interpolation, an algebraic galactic force, an open-system source and multiple cosmological claims. Subsequent equation audits and numerical gates showed that these ingredients do not yet follow from one action. The purpose of this alpha is therefore architectural: it states what the model is, which results are presently supportable, and what must still be calculated.

Four labels are used throughout:

Derived. A checked result that follows from declared assumptions within its stated domain.

Conditional. A result that also requires a named postulate, matching hypothesis, background choice or symmetry limit.

Open. A well-defined derivation or numerical test remains incomplete.

Rejected. The stated derivation or implementation failed and is not live support for the framework.

These labels apply to derivations rather than to broad themes. For example, topology-dependent Casimir stress is derived for the declared periodic scalar calculation, while a universal 13/12 expansion ratio from cycle counting is rejected. A rejected shortcut does not reject T^3 topology; likewise, a successful local effective action does not establish a relativistic or cosmological completion.

1.1 Conventions

We use metric signature $(-,+,+,+)$ and natural units $c = \hbar = 1$ except where factors are restored explicitly, as in the Casimir lattice energy. The reduced Planck mass and gravitational coupling satisfy

$$M_{\mathrm{P}}^{-2} = 8\pi G, \qquad \kappa = 8\pi G. \tag{1}$$

These conventions make the covariant $Y^{3/2}$ coefficient and the static cubic-gradient coefficient directly comparable.

The reconstruction rule is that no observational claim may have a stronger status than its upstream field sector. Historical calculations remain useful as diagnostics and provenance, but they are not promoted into the core merely because their numerical values are suggestive.

2 Core fields and layered architecture

2.1 Metric, condensate, and finite-density frame

The gravitational variable is the spacetime metric $g_{\mu\nu}$. The candidate plenum order parameter is complex,

$$\Phi = \frac{\rho}{\sqrt{2}} e^{i\Theta}, \tag{2}$$

so that amplitude fluctuations, phase excitations, zeros of the order parameter and winding sectors can be represented within one microscopic language. UVIR-001 rejects the minimally kinetic analytic candidate as the direct origin of the spatial cubic force operator, but it does not remove these density, circulation, defect and topology roles.

A unit timelike vector U^μ defines the preferred frame,

$$U^\mu U_\mu = -1, \qquad h^{\mu\nu} = g^{\mu\nu} + U^\mu U^\nu. \tag{3}$$

UVIR-003 Stage A provisionally chooses U^μ as an independent constrained dynamical field. The condensate current may be aligned with it through an interaction energy, but is not algebraically identified with it. This avoids double-counting a derived condensate velocity and an independent aether field. The choice remains provisional until the coupled constraint and stability gate closes.

Table 1: High-level status at the start of the core recovery.

Component	Status	Boundary of the result
Finite-density plenum	Conditional	Physical postulate; microscopic action open
Compact T^3 topology	Conditional	Adopted boundary condition; scale open
$Y^{3/2}$ square-root branch	Conditional	Local, quasi-static, high symmetry
Born–Infeld origin	Rejected	Does not give the required IR response
$C_{\mathrm{proj}} = 2/3$ local vertex	Conditional	Geometric motivation; matching open
Rectangular-T^3 Casimir tensor	Derived	Periodic free scalar with stated moduli
Persistent 13/12 from free-scalar Casimir stress	Rejected	CBR-001 finds only transient crossings
Persistent 13/12 from a derived driven ITSM sector	Open	Requires TOP/WAK/reservoir closure and CBR-002
Screening, PPN and lensing	Open	Relativistic completion absent
Scalar metric–aether–condensate block	Derived (bounded)	Finite-q pass; homogeneous gauge orbit removed
Aether vertices	Derived (bounded)	3D quartic; COM s-channel gauge hold
Constrained nonlinear scalar action	Derived (bounded)	Complete finite-q S_2, $L_4[x, z_1]$ and regular basis; projected amplitude open
Realistic disk field solution	Open	Periodic nonlinear PDE required
Driven reservoir anisotropy	Open	Constitutive conversion not derived

2.2 Separate preferred-frame force scalar and matter

The low-acceleration force mediator is denoted ψ. Under the selected UVIR-002 architecture it is a separate preferred-frame scalar, not the phase of Φ relabelled at low energy. Independent temporal and spatial invariants can therefore be formed from $U^\mu \nabla_\mu \psi$ and $h^{\mu\nu} \nabla_\mu \psi \nabla_\nu \psi$. The cubic spatial operator is a conditional infrared EFT input; it has not been matched to a microscopic completion.

Matter fields are collected in Ψ_m. The matter–force interaction must be derived once, then used consistently in both the scalar source and the matter stress-energy exchange. No coefficient inferred from a convenient normalization is promoted to an observable coupling before MAT-001.

2.3 Topology, reservoir, and non-equilibrium sectors

Spatial slices are assigned compact T^3 boundary conditions with lengths L_i. Their ratios are shape moduli. The global topology controls allowed modes and winding; it does not directly fix every local Wilson coefficient.

An explicit reservoir sector R permits the observable matter–plenum system to be open while the complete covariant system remains conserved. A wake or memory variable W and a shape sector are permitted research sectors, not assumed completed degrees of freedom.

The schematic inventory is

$$S_{\mathrm{complete}} = S_{\mathrm{EH}} + S_\Phi + S_U + S_{\mathrm{align}} + S_\psi + S_m + S_R + S_{\mathrm{int}} + S_W, \tag{4}$$

where S_W is included only if a causal and stable wake description is found. The displayed sectors define a research architecture, not a completed microscopic action. Effective operators must be matched and overlapping UV and IR descriptions must not be counted as independent copies of the same degree of freedom.

2.4 Architecture versus prediction

Equation (4) is a sector map. In particular,

$$\text{energy throughput} \not\Rightarrow \text{anisotropic pressure} \tag{5}$$

without a shape-dependent conversion law, and

$$\text{compact topology} \not\Rightarrow C_{\text{proj}} = \frac{2}{3} \quad \text{or} \quad \frac{H_t}{H_p} = \frac{13}{12} \tag{6}$$

without separate matching and backreaction calculations. Likewise, a positive frozen-background dispersion is not a proof of coupled stability, causality or weak coupling.

3 Ultraviolet–infrared hierarchy

3.1 UVIR-001: the minimal condensate route closes negative

The tested microscopic candidate is

$$S_\Phi = \int \mathrm{d}^4x\sqrt{-g}\left[-\nabla_\mu\Phi^*\nabla^\mu\Phi - m^2|\Phi|^2 - \frac{\lambda_4}{2}|\Phi|^4 - \frac{\lambda_6}{3\Lambda^2}|\Phi|^6\right]. \tag{7}$$

Writing $\Phi = \rho e^{i\Theta}/\sqrt{2}$ and $Z = -\nabla_\mu\Theta\nabla^\mu\Theta$, its stable finite-density branch can be reduced to a phase action $P(Z)$. In the pure-sextic limit the explicit amplitude integration gives

$$P(Z) = \frac{2\Lambda}{3\sqrt{\lambda_6}}(Z - m^2)^{3/2}. \tag{8}$$

This nonanalytic power does not supply the required static spatial operator. About a timelike background $Z_0 = \mu^2$, a static phase gradient $q = |\nabla\vartheta|$ instead yields

$$P(\mu^2 - q^2) - P(\mu^2) = -P_Z(\mu^2)q^2 + \mathcal{O}(q^4), \qquad P_Z = \frac{\rho_0^2}{2} > 0. \tag{9}$$

The leading response is therefore quadratic for every stable nonzero branch with the canonical phase kinetic term. UVIR-001 is closed negative for this minimal candidate. The result rejects neither the condensate's density and topology roles nor a bottom-up spatial $Y^{3/2}$ EFT.

3.2 UVIR-002: provisional two-sector route

UVIR-002 also rejects a standalone nonrelativistic gradient-dominated branch: the branch producing the desired cubic spatial action has a negative quadratic time kinetic coefficient. The least-incomplete route tested separates the jobs of the condensate and force mediator. The independent invariants are

$$\mathcal{Q} = U^\mu\nabla_\mu\psi, \qquad \mathcal{Y} = h^{\mu\nu}\nabla_\mu\psi\nabla_\nu\psi. \tag{10}$$

Here Φ retains finite density, circulation, defects and topology, while (ψ, U^μ) carries the conditional low-acceleration response. This is a provisional architecture selection, not microscopic closure.

3.3 UVIR-003 Stage A action and limited pass

Stage A chooses an independent unit vector and retains the complete Einstein–aether two-derivative basis,

$$\begin{aligned}\mathcal{L}_U = -\frac{M_U^2}{2}\big[&c_1(\nabla_\mu U_\nu)(\nabla^\mu U^\nu) + c_2(\nabla_\mu U^\mu)^2 \\ &+ c_3(\nabla_\mu U_\nu)(\nabla^\nu U^\mu) - c_4 a_\mu a^\mu\big] + \frac{\lambda}{2}(U^\mu U_\mu + 1),\end{aligned} \tag{11}$$

where $a_\mu = U^\nu \nabla_\nu U_\mu$. A positive alignment energy couples the frame to the condensate current without imposing equality. The regulated force truncation is

$$\mathcal{L}_\psi = \frac{K_Q}{2}\mathcal{Q}^2 - A\mathcal{Y}^{3/2} - \frac{\gamma}{2M_*^2}(\Delta_U \psi)^2. \tag{12}$$

The symbolic Stage-A calculation gives the necessary frozen-metric conditions

$$c_{14} > 0, \qquad c_1 > 0, \qquad c_{123} > 0, \qquad K_Q > 0, \quad A > 0, \quad \gamma > 0, \tag{13}$$

and at zero background gradient the regulator produces

$$\omega^2 = \frac{\gamma}{K_Q M_*^2} k^4. \tag{14}$$

This is a pass only in the declared flat, constant-frame decoupling limit.

3.4 Stage B frame-sector diagnostic

A verified sign map to the Einstein–aether review [3] preserves all four operator labels. Its normalization also requires

$$\alpha_i = r_U c_i, \qquad r_U = \frac{M_U^2}{M_P^2}, \tag{15}$$

because the source places the aether operators under the Einstein–Hilbert prefactor, while the Stage-A action declares an independent M_U. The published coupled metric–aether result therefore gives

$$s_2^2 = \frac{1}{1-\alpha_{13}}, \tag{16}$$

$$s_1^2 = \frac{\alpha_1 - \frac{1}{2}\alpha_1^2 + \frac{1}{2}\alpha_3^2}{\alpha_{14}(1-\alpha_{13})}, \tag{17}$$

$$s_0^2 = \frac{\alpha_{123}(2-\alpha_{14})}{\alpha_{14}(1-\alpha_{13})(2+\alpha_{13}+3\alpha_2)}. \tag{18}$$

As $r_U \to 0$ at fixed c_i ratios, the spin-1 and spin-0 expressions reduce to the Stage-A ratios c_1/c_{14} and c_{123}/c_{14}. This confirms that those ratios are weak-metric-coupling limits, not the full mode speeds. It is a checked literature substitution, not an independent ITSM-side linearized Einstein-equation derivation, and it excludes the coupled Φ–U–ψ system.

3.5 Zero-gradient force-block factorization

On the declared background $\bar{\psi} =$ constant, every derivative of the force field begins at first perturbative order. Consequently,

$$\mathcal{Q}^2 = \dot{\pi}^2 + \mathcal{O}(3), \qquad \mathcal{Y}^{3/2} = \mathcal{O}(3), \qquad (\Delta_U \psi)^2 = (\nabla^2 \pi)^2 + \mathcal{O}(3), \tag{19}$$

where metric and frame corrections enter only in the indicated higher-order terms. The force scalar therefore factorizes from the lapse, shift, frame and condensate blocks at quadratic order for the declared truncation:

$$S_\pi^{(2)} = \frac{1}{2}\int \mathrm{d}^4x \left[K_Q \dot{\pi}^2 - \frac{\gamma}{M_*^2}(\nabla^2\pi)^2 \right]. \tag{20}$$

It carries one nonnegative $z = 2$ scalar mode for $K_Q, \gamma > 0$. This bounded result does not reduce the remaining metric–frame–condensate block.

A constant field redefinition $\psi_c = s\psi$ sends

$$K_Q \to \frac{K_Q}{s^2}, \quad A \to \frac{A}{s^3}, \quad \gamma \to \frac{\gamma}{s^2}, \quad C_m \to \frac{C_m}{s}, \quad q \to sq. \tag{21}$$

Thus γ/K_Q, $A/K_Q^{3/2}$, $C_m/\sqrt{K_Q}$ and Aq/K_Q are invariant, whereas K_Q alone is not. UVIR-003 can determine a structural stability domain in invariant ratios, but a parent microscopic calculation or MAT-001 must fix the physical normalization and select a point in that domain. The earlier $K_Q \sim M_P^2$ estimate remains a speculative dimensional analogy.

3.6 Scalar ADM readiness

The intended coupled reduction was to be performed around Minkowski space with $\Phi = \rho_0 e^{i\mu t}/\sqrt{2}$, constant U^μ and constant $\bar{\psi}$. This background is not on shell for the declared gravitational action. Writing $s = \rho_0^2$, the homogeneous condensate has

$$p_\Phi = \frac{1}{2}s\mu^2 - V(\rho_0), \qquad \rho_\Phi = \frac{1}{2}s\mu^2 + V(\rho_0), \qquad \rho_\Phi + p_\Phi = s\mu^2 > 0. \tag{22}$$

A constant vacuum-energy subtraction shifts ρ and p oppositely, so it cannot cancel their sum. It therefore cannot make both flat-background Einstein equations vanish on the nonzero finite-density branch.

An exact stationary completion would need a reservoir or driver satisfying

$$\rho_R = -\rho_\Phi, \qquad p_R = -p_\Phi, \qquad \rho_R + p_R = -s\mu^2. \tag{23}$$

This is non-vacuum stress. Its scalar perturbations generally enter the lapse and shift constraints and cannot be silently omitted. Until a covariant support action, a controlled rigid-support limit, or a consistent cosmological background is declared, eliminating the ADM constraints would produce an off-shell kinetic matrix. The scalar reduction is therefore blocked at readiness rather than counted as a stability failure. The zero-gradient force factorization remains valid because that block is quadratically independent on the declared constant-force background.

3.7 Background-completion screen

For a minimally coupled shift-symmetric support scalar with $X_R = -\nabla_\mu\chi\nabla^\mu\chi/2 > 0$ and $\mathcal{L}_R = P_R(X_R)$,

$$\rho_R + p_R = 2X_R P_{R,X}. \tag{24}$$

Exact Minkowski support would therefore require

$$P_{R,X} = -\frac{s\mu^2}{2X_R} < 0. \tag{25}$$

The principal quadratic fluctuation action is

$$\mathcal{L}_R^{(2)} = \frac{1}{2}(P_{R,X} + 2X_R P_{R,XX})\dot{\pi}_R^2 - \frac{1}{2}P_{R,X}(\nabla\pi_R)^2. \tag{26}$$

A healthy two-derivative short-wavelength scalar requires both $P_{R,X} + 2X_R P_{R,XX} > 0$ and $P_{R,X} > 0$, in conflict with Eq. (25). This is the local version of the familiar instability obstruction for simple phantom-like scalar backgrounds [7]. The ghost-condensate point has $P_{R,X} = 0$ and hence zero enthalpy; its k^4 excitation does not by itself supply the missing background stress [8].

Stable NEC violation can be engineered with additional EFT structure [9], but importing that structure would add a new microscopic sector, cutoff and stability gate. A prescribed rigid stress is also insufficient for the full problem because it has no action-derived lapse or shift response. The screen therefore selects no new exact-Minkowski support field.

The least-assumptive route is a self-consistent evolving flat-FRW background,

$$M_{\cos}^2 = M_P^2 + \frac{M_U^2}{2}(c_1 + 3c_2 + c_3), \tag{27}$$

$$3M_{\cos}^2 H^2 = \rho_{\text{total}}, \qquad -2M_{\cos}^2 \dot{H} = \rho_{\text{total}} + p_{\text{total}}. \tag{28}$$

For isolated condensate charge $n = \mu\rho^2$,

$$\dot{n} + 3Hn = 0. \tag{29}$$

Thus the amplitude or chemical potential generally evolves. Exact constant n in expansion would require a separately declared charge source $S_N = 3Hn$; the stress-energy exchange vector Q^ν_{syn} is not automatically that source.

3.8 Evolving FRW existence check

For the comoving unit aether, the Stage-A contractions reduce to $3H_N^2$, $9H_N^2$, $3H_N^2$ and zero acceleration squared. The alignment projector annihilates the parallel homogeneous condensate current, and the constant force field has no background derivatives. The reduced action is

$$S_{\text{bg}} = \int \mathrm{d}t \left[-\frac{3M_{\cos}^2 a\dot{a}^2}{N} + \frac{a^3}{2N}(\dot{\rho}^2 + \rho^2\dot{\Theta}^2) - Na^3V(\rho) \right]. \tag{30}$$

Its amplitude and phase equations are

$$\ddot{\rho} + 3H\dot{\rho} - \rho\mu^2 + V_{,\rho} = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(a^3\rho^2\mu) = 0. \tag{31}$$

A representative dimensionless integration, not a parameter fit, uses $M_P = 1$, $M_U = 0.5$, $(c_1, c_2, c_3, c_4) = (0.10, 0.05, 0.05, 0.05)$ and positive mass–quartic–sextic couplings. From $t = 0$ to 8, it remains regular and expanding:

$$(a, \rho, \mu, H) : (1, 1, 1.123610, 0.619841) \longrightarrow (4.571577, 0.134236, 0.652648, 0.076405). \tag{32}$$

The maximum relative Friedmann residual is 2.124×10^{-10}, charge drift is 2.220×10^{-16} and the continuity residual is 1.898×10^{-15}. This verifies an on-shell existence branch and removes the background blocker. It does not select a fiducial cosmology or establish perturbation stability.

The metric–frame–condensate constraint reduction is now ready to begin on this time-dependent background. A subhorizon limit can test local modes only for $k/a \gg H$; the strict low-k question remains part of the complete cosmological perturbation system.

3.9 Scalar ADM principal-symbol reduction

The first scalar constraint subgate uses aether-unitary gauge,

$$N = 1 + \delta N, \qquad N_i = \partial_i \beta, \qquad h_{ij} = a^2 e^{2\mathcal{R}} \delta_{ij}, \qquad U^\mu = n^\mu, \tag{33}$$

and freezes the evolving background coefficients over a wavelength. For $q_{\rm phys} = k/a \gg H$, the principal lapse and scalar-shift solutions are

$$\delta N = -\frac{2M_P^2}{M_U^2 c_{14}} \mathcal{R} + \frac{\dot\rho\, \delta\dot\rho + \rho^2 \mu \dot\vartheta}{M_U^2 c_{14} q_{\rm phys}^2}, \tag{34}$$

$$\beta = -\frac{2M_{\rm cos}^2 \dot{\mathcal{R}} + \dot\rho\, \delta\rho + \rho^2 \mu \vartheta}{M_U^2 c_{123} q_{\rm phys}^2}. \tag{35}$$

With $F = M_{\rm cos}^2/M_P^2 = 1 + (\alpha_{13} + 3\alpha_2)/2$, their elimination gives

$$\mathcal{L}_{\mathcal{R}}^{(2)} = \mathcal{K}_{\mathcal{R}} \dot{\mathcal{R}}^2 - \mathcal{G}_{\mathcal{R}} q_{\rm phys}^2 \mathcal{R}^2, \quad \mathcal{K}_{\mathcal{R}} = \frac{2M_P^2 F (1 - \alpha_{13})}{\alpha_{123}}, \quad \mathcal{G}_{\mathcal{R}} = \frac{M_P^2 (2 - \alpha_{14})}{\alpha_{14}}. \tag{36}$$

Their ratio independently recovers the exact coupled Einstein–aether spin-0 speed [3],

$$s_0^2 = \frac{\alpha_{123}(2 - \alpha_{14})}{\alpha_{14}(1 - \alpha_{13})(2 + \alpha_{13} + 3\alpha_2)}. \tag{37}$$

The condensate velocity Hessian retains a leading lapse-induced correction whose determinant factorizes as

$$\det \mathbf{K}_\Phi(q_{\rm phys}) = \rho^2 \left[1 - \frac{\dot\rho^2 + \rho^2 \mu^2}{M_U^2 c_{14} q_{\rm phys}^2}\right]. \tag{38}$$

Thus the controlled domain is not merely $q_{\rm phys} \gg H$ but

$$q_{\rm phys} \gg \max(H, q_{\rm ADM}), \qquad q_{\rm ADM}^2 = \frac{\dot\rho^2 + \rho^2 \mu^2}{M_U^2 c_{14}}. \tag{39}$$

At the representative dimensionless point, $\mathcal{K}_{\mathcal{R}} = 39.94375$, $\mathcal{G}_{\mathcal{R}} = 52.33333$ and $s_0^2 = 1.31018$. The principal block is positive, but its cone is wider than the metric cone. This is a bounded subhorizon result with a multicone-causality flag, not a full finite-k or low-k cosmological stability proof.

3.10 Time-dependent finite-wavenumber scalar reduction

The complete quadratic scalar reduction retains the evolving background and all terms through the homogeneous order. Define

$$\Sigma = q_{\rm phys}^2 \beta, \qquad D_{123} = M_U^2 c_{123}, \qquad C_{14} = M_U^2 c_{14}, \qquad M = M_{\rm cos}^2. \tag{40}$$

For every $q_{\rm phys} > 0$, the lapse and momentum constraints can be written

$$\mathcal{L}^{(2)} = \mathcal{L}_0 + \mathbf{z}^\mathsf{T} \mathbf{J} + \frac{1}{2} \mathbf{z}^\mathsf{T} \mathbf{C} \mathbf{z}, \qquad \mathbf{z} = (\delta N, \Sigma)^\mathsf{T}, \tag{41}$$

with

$$\mathbf{C} = \begin{pmatrix} C_{14} q_{\rm phys}^2 - 2V & 2MH \\ 2MH & -D_{123} \end{pmatrix}. \tag{42}$$

The source vector retains H, $\dot\rho$, μ and the amplitude-potential terms. Exact elimination gives

$$\mathbf{z} = -\mathbf{C}^{-1}\mathbf{J}, \qquad \mathcal{L}^{(2)}_{\rm red} = \mathcal{L}_0 - \frac{1}{2}\mathbf{J}^{\mathsf{T}}\mathbf{C}^{-1}\mathbf{J}. \tag{43}$$

Coefficient derivatives in the reduced equations include $\dot H$, $\ddot\rho$, $\dot\mu$ and $\dot q_{\rm phys} = -Hq_{\rm phys}$ for a fixed comoving mode.

On the background equations, the exact reduced velocity-Hessian determinant is

$$\det \mathbf{K}_{\rm red} = \frac{2M\rho^2(2M - 3D_{123})C_{14}q^2_{\rm phys}}{C_{14}D_{123}q^2_{\rm phys} - 2D_{123}V + 4M^2H^2}. \tag{44}$$

Its high-wavenumber limit reproduces the principal curvature and condensate blocks above. A scan of 801 background times and 61 logarithmic wavenumbers, $10^{-3} \le q_{\rm phys}/H \le 10^3$, finds three positive and no negative kinetic eigenvalues at all 48,861 nonzero-wavenumber samples. The constraint matrix remains nonsingular; its smallest sampled singular value is 0.0501148.

Equation (44) also exposes a homogeneous rank change. The smallest normalized eigenvalue scales numerically as $q^{2.00066}_{\rm phys}$, and the kinetic rank falls from three to two at $q_{\rm phys} = 0$. The follow-on audit evaluates the endpoint exactly and finds

$$\mathbf{K}_{\rm red}(0)\begin{pmatrix} H & \dot\rho & \mu \end{pmatrix}^{\mathsf{T}} = 0. \tag{45}$$

This vector is tangent to $(\ln a, \rho, \Theta)$ under a homogeneous time translation. The two combinations

$$Q_\rho = \delta\rho - \frac{\dot\rho}{H}\mathcal{R}, \qquad Q_\Theta = \vartheta - \frac{\mu}{H}\mathcal{R} \tag{46}$$

annihilate that orbit. After phase normalization by ρ, their physical $q = 0$ kinetic block has two positive and no negative eigenvalues at every one of the 801 representative trajectory points; its minimum eigenvalue is 0.937286. At $q_{\rm phys}/H = 10^{-3}$ the smallest finite-q eigenvector has minimum alignment cosine 0.9999999999999994 with the time-shift orbit. The rank loss is therefore a gauge endpoint, not a third physical homogeneous scalar. Canonically normalizing its vanishing norm would manufacture a gauge-dependent strong-coupling scale.

3.11 Bounded khronon cubic readiness result

As a controlled next step, use the hypersurface-orthogonal aether–khronon equivalence [4] and restore the preferred-time scalar through

$$T = t + \pi, \qquad U_\mu = -\frac{\partial_\mu T}{\sqrt{-(\partial T)^2}}. \tag{47}$$

For a one-dimensional longitudinal profile with the metric held flat, the four-operator aether action gives, after suppressing the overall M_U^2 factor,

$$\mathcal{L}_2 = \frac{1}{2}\left[c_{14}\pi^2_{tx} - c_{123}\pi^2_{xx}\right], \tag{48}$$

$$\begin{aligned} \mathcal{L}_3 = &- c_{14}\pi_t\pi^2_{tx} + c_{123}\pi_t\pi^2_{xx} - c_{14}\pi_{tt}\pi_{tx}\pi_x \\ &+ (2c_{123} - c_{14})\pi_{tx}\pi_x\pi_{xx}. \end{aligned} \tag{49}$$

For every nonzero Fourier mode, $\chi_k = M_U\sqrt{c_{14}}|k|\pi_k$ supplies the quadratic canonical normalization and $c_s^2 = c_{123}/c_{14}$. This longitudinal result is a one-dimensional check on the complete three-dimensional basis below.

3.12 Three-dimensional cubic and constraint-order result

Let $p_i = \partial_i \pi$, $v_i = \partial_i \dot{\pi}$, $H_{ij} = \partial_i \partial_j \pi$ and $H = \Delta \pi$. With the overall M_U^2 factor suppressed, direct expansion of the hypersurface-orthogonal aether action in flat decoupling gives

$$\mathcal{L}_2 = \frac{1}{2}\left[c_{14} v_i v_i - (c_1 + c_3) H_{ij} H_{ij} - c_2 H^2\right], \tag{50}$$

$$\begin{aligned}\mathcal{L}_3 = & - c_{14} \left(\ddot{\pi}\, p_i v_i + \dot{\pi}\, v_i v_i\right) + (c_1 + 2c_3 - c_4) p_i H_{ij} v_j + 2c_2 H p_i v_i \\ & + (c_1 + c_3) \dot{\pi}\, H_{ij} H_{ij} + c_2 \dot{\pi}\, H^2. \end{aligned} \tag{51}$$

Suppressing transverse derivatives reduces this expression exactly to Eqs. (48)–(49).

The perturbative constraint order can also be fixed without guessing the second-order lapse or shift. For nondynamical variables z with

$$\mathcal{L}_2 = \mathcal{L}_{2,\mathrm{phys}} + z^{\mathsf{T}} J + \frac{1}{2} z^{\mathsf{T}} C z, \qquad z_1 = -C^{-1} J, \tag{52}$$

substitution of $z = z_1 + z_2 + \cdots$ makes the term linear in z_2 vanish by $C z_1 + J = 0$; the remaining quadratic term starts at fourth order. Replacing z_1 by $z_1 + z_2$ inside $\mathcal{L}_3$ likewise first changes quartic order. Hence the reduced cubic action requires the first-order constraints, not explicit second-order lapse and shift solutions.

A representative operator-basis estimate gives

$$q_{\mathrm{NDA}} = 0.125778823734, \qquad 0.2029209475 \leq \frac{q_{\mathrm{NDA}}}{H} \leq 1.6462061767 \tag{53}$$

along the sampled trajectory. These values are not a physical cutoff: they change under field redefinitions and omit cosmological mixing. Moreover, for linear dispersion $|\omega| = c_s |\mathbf{k}|$, on-shell three-point conservation saturates the triangle inequality and forces the momenta to be collinear. A non-collinear on-shell cubic three-point amplitude therefore cannot define the desired interaction scale. The next invariant calculation is the complete constrained scalar $2 \to 2$ amplitude, including cubic exchange, the quartic contact vertex, physical eigenmode projection and a stated unitarity criterion. The action-dependent warnings and healthy-extension results in Refs. [5, 6] reinforce this boundary. UVIR-003 remains open.

3.13 Quartic contact and exchange readiness result

The normalized khronon expansion has also been carried through fourth order in three spatial dimensions. The expanded flat-decoupling action contains 96 quartic monomials, reproduces the quadratic and cubic results above and reduces exactly to an independently generated longitudinal quartic action. For unit external momentum in elastic centre-of-mass kinematics, its on-shell contact coefficient reduces to

$$C_4(\theta) = 4\left[\frac{c_{123}^2}{c_{14}} - (2c_{123} - c_{14}) \cos^2 \theta\right]. \tag{54}$$

This is a contact coefficient, not a partial-wave amplitude or a cutoff.

For static momentum transfer, the polarized cubic vertex factorizes by $x^2 + y^2 - 1$ where $(y, 0, x)$ is the outgoing unit direction. Hence elastic t/u cubic exchange vanishes exactly. The centre-of-mass s channel instead has internal momentum $q_s = (-2\omega, \mathbf{0})$. Since the flat khronon inverse kernel is

$$D_\pi(q) = M_U^2 c_{14} |\mathbf{q}|^2 \left(q_0^2 - c_s^2 |\mathbf{q}|^2\right), \tag{55}$$

it cannot be inverted on that homogeneous gauge orbit. Dropping the channel would not produce a physical amplitude.

The required quartic constraint order is also fixed. If J_2 denotes the quadratic source for the second-order lapse and shift solution, then

$$z_2 = -C^{-1} J_2, \qquad \mathcal{L}^{(4)}_{\text{red}} = \mathcal{L}_4[x, z_1] - \frac{1}{2} J_2^{\mathsf{T}} C^{-1} J_2. \tag{56}$$

Third-order constraint solutions cancel at this order. The next invariant calculation is therefore the evolving-FRW cubic and quartic action with this Schur complement, followed by physical scalar projection and a gauge-regular $2 \to 2$ unitarity amplitude. The flat quartic basis passes as a readiness result, while the physical scale and UVIR-003 remain open.

3.14 Nonlinear ADM action provenance

Before computing the cosmological quadratic constraint source J_2, the nonlinear parent action must be fixed rather than inferred from the reduced quadratic system. In aether-unitary gauge the exact two-derivative $g + U + \Phi$ ADM density is

$$\begin{aligned} \mathcal{L}_{gU\Phi} = \frac{N\sqrt{h}}{2} \big[& M_P^2 {}^{(3)}R + (M_P^2 - M_U^2 c_{13}) K_{ij} K^{ij} - (M_P^2 + M_U^2 c_2) K^2 + M_U^2 c_{14} a_i a^i \\ & + n(\varrho)^2 + \varrho^2 n(\theta)^2 - h^{ij} \{ D_i \varrho D_j \varrho + \varrho^2 D_i \theta D_j \theta \} - 2V(\varrho) \big]. \end{aligned} \tag{57}$$

Since $J^\mu_\Phi = -\varrho^2 \nabla^\mu \theta$, the alignment term becomes

$$\mathcal{L}_{\text{align}} = -\frac{N\sqrt{h}}{2} \zeta_{\text{align}} \varrho^4 h^{ij} D_i \theta D_j \theta. \tag{58}$$

Symbolic expansion of this parent block exactly reproduces the FRW minisuperspace action and the complete finite-q quadratic constraint matrix and source already reported. Thus the quadratic $g + U + \Phi$+alignment system has verified nonlinear action provenance.

At the preceding action-provenance checkpoint, the complete force sector was not yet fixed: Stage A had defined Δ_U only in the constant hypersurface-orthogonal frame. A separate issue remains at the selected zero-gradient background, where $Y = \epsilon^2 Y_2$ gives

$$Y^{3/2} = |\epsilon|^3 Y_2^{3/2}. \tag{59}$$

This is even under $\epsilon \to -\epsilon$, whereas a nonzero homogeneous cubic Taylor polynomial is odd. Consequently an ordinary analytic cubic vertex is not defined there. Track A resolves the action-definition hold below while retaining this exact non-analytic functional; it does not turn the local force interaction into a homogeneous perturbative amplitude.

A bounded completion audit identifies the rest-space Laplacian

$$\Delta_U \psi := D_\mu D^\mu \psi = h^{\mu\nu} \nabla_\mu \nabla_\nu \psi + \Theta_U \mathcal{Q}, \qquad \Theta_U := \nabla_\mu U^\mu. \tag{60}$$

as the minimal evolving-frame operator. Track A now adopts Eq. (60), retains exact $Y^{3/2}$ and assigns its ordinary perturbative force analysis to a declared local nonzero-gradient background. No smoothing scale or canonical linear-Y term is introduced.

On the homogeneous zero-gradient FRW branch, with $N_i = \partial_i \beta$, $h_{ij} = a^2 e^{2\mathcal{R}} \delta_{ij}$ and $\psi = \bar{\psi} + \pi$, the adopted operator is

$$D_\mu D^\mu \psi = a^{-2} e^{-2\mathcal{R}} (\partial^2 \pi + \partial_i \mathcal{R}\, \partial_i \pi). \tag{61}$$

Its exact symbolic expansion, together with $\mathcal{Q}^2$ and the retained IR functional, reproduces the quadratic $z = 2$ block and is verified through direct quartic order. Before finite-q scalar-shift normalization, the force contribution to the quadratic constraint source is

$$\begin{aligned} J^{(\psi)}_{2,\delta N} &= -\frac{a^3 K_Q}{2}\dot{\pi}^2 - \frac{\gamma}{2M_*^2 a}(\partial^2\pi)^2, \\ J^{(\psi)}_{2,\beta} &= aK_Q\partial_i(\dot{\pi}\,\partial_i\pi). \end{aligned} \tag{62}$$

The regulator has no shift source. Exact $Y^{3/2}$ is constraint independent at cubic amplitude order on the zero-gradient branch, so its contribution to J_2 there is zero; this does not make it an analytic three-point vertex. The force component passes and can therefore be combined with the fixed nonlinear parent block.

3.15 Origin-linear and dressed finite-wavenumber sources

Let $M = M^2_{\rm cos}$ and let D_i denote the physical derivative on the unperturbed FRW leaf. The coefficient linear in the constraints at their origin is a verified component of the second-order source. Its lapse part is

$$\begin{aligned} J^{\rm origin}_{2,N} = 3M\dot{\mathcal{R}}^2 &+ 18MH\mathcal{R}\dot{\mathcal{R}} - 2M_P^2\mathcal{R}D^2\mathcal{R} - M_P^2 D_i\mathcal{R}D_i\mathcal{R} \\ &- \frac{1}{2}\left(\dot{\delta\varrho}^2 + \mu^2\delta\varrho^2 + 4\varrho\mu\delta\varrho\dot{\vartheta} + \varrho^2\dot{\vartheta}^2 + V_{\varrho\varrho}\delta\varrho^2\right) \\ &- 3\mathcal{R}\left(\dot{\varrho}\dot{\delta\varrho} + \varrho\mu^2\delta\varrho + \varrho^2\mu\dot{\vartheta} + V_\varrho\delta\varrho\right) \\ &- \frac{1}{2}\left[D_i\delta\varrho D_i\delta\varrho + \varrho^2(1+\zeta_{\rm align}\varrho^2)D_i\vartheta D_i\vartheta\right] \\ &- \frac{K_Q}{2}\dot{\pi}^2 - \frac{\gamma}{2M_*^2}(D^2\pi)^2. \end{aligned} \tag{63}$$

With $\Sigma = -D^2\beta = q^2_{\rm phys}\beta$, the shift component is

$$J^{\rm origin}_{2,\Sigma} = -2M\left(\mathcal{R}\dot{\mathcal{R}} + \frac{1}{2}H\mathcal{R}^2\right) + (D^2)^{-1}D_iW_i, \tag{64}$$

$$\begin{aligned} W_i = 2M(\dot{\mathcal{R}} + H\mathcal{R})D_i\mathcal{R} &- \dot{\delta\varrho}D_i\delta\varrho - \mathcal{R}\dot{\varrho}D_i\delta\varrho \\ &- \varrho^2\dot{\vartheta}D_i\vartheta - 2\varrho\mu\delta\varrho D_i\vartheta - \mathcal{R}\varrho^2\mu D_i\vartheta - K_Q\dot{\pi}D_i\pi. \end{aligned} \tag{65}$$

The inverse Laplacian in Eq. (64) is defined only at $q_{\rm phys} > 0$, the domain in which Σ is an independent constraint; it does not alter the homogeneous gauge-orbit result.

The origin-linear coefficient is not the complete source because the cubic ADM action contains nonlinear lapse and scalar-shift dependence. With $z_1 = -C^{-1}J_1$, the correct functional is

$$S_2 = \left.\frac{\partial L_3[x,z]}{\partial z}\right|_{z=z_1} = J_2^{\rm origin} + \Delta S_2^{(g+U)} + \Delta S_2^{(\Phi)}. \tag{66}$$

The generic gravity–aether term retains all Hessian, $D_i\mathcal{R}D_i\beta$ and lapse-gradient contractions. The condensate term includes temporal lapse and shift advection. The Track-A force cubic block is affine in the constraints, so it adds no nonlinear correction beyond $J_2^{\rm origin}$ on the declared homogeneous zero-gradient branch.

The complete generic fourth-order contact functional has also been expanded at z_1. Eliminating the second-order constraints therefore gives

$$\mathcal{L}_{\rm red}^{(4)} = L_4[x, z_1] - \frac{1}{2} S_2^{\mathsf{T}} C^{-1} S_2. \tag{67}$$

This supersedes the provisional expression using $J_2^{\rm origin}$ alone. The complete S_2, generic $L_4[x, z_1]$ contact and reduced quartic functional pass at $q_{\rm phys} > 0$. This is still an unprojected functional, not a physical scattering amplitude.

3.16 Regular finite-wavenumber physical basis

The homogeneous time-shift orbit can be separated from the finite-wavenumber preferred-time gradient by defining

$$\Xi = \frac{q_{\rm phys}}{H}\mathcal{R}, \qquad Q_\rho = \delta\rho - \frac{\dot\rho}{H}\mathcal{R}, \qquad Q_\chi = \rho\left(\vartheta - \frac{\mu}{H}\mathcal{R}\right). \tag{68}$$

For $q_{\rm phys} > 0$ the inverse map is

$$\mathcal{R} = \frac{H}{q_{\rm phys}}\Xi, \quad \delta\rho = Q_\rho + \frac{\dot\rho}{q_{\rm phys}}\Xi, \quad \vartheta = \frac{Q_\chi}{\rho} + \frac{\mu}{q_{\rm phys}}\Xi. \tag{69}$$

The first column is the time-shift vector $(H, \dot\rho, \mu)^{\mathsf{T}}/q_{\rm phys}$. It converts the verified $q_{\rm phys}^2$ kinetic collapse into a finite gradient-mode normalization without introducing an exactly homogeneous Ξ degree of freedom.

If T denotes Eq. (69), the transformed determinant is

$$\det K_{\rm phys} = \frac{2MH^2(2M - 3D_{123})C_{14}}{C_{14}D_{123}q_{\rm phys}^2 - 2D_{123}V + 4M^2H^2}. \tag{70}$$

It has a finite nonzero on-shell limit as $q_{\rm phys} \to 0$. A scan of 801 background times and 49 logarithmic wavenumbers over $10^{-3} \le q_{\rm phys}/H \le 10^3$ finds three positive and no negative eigenvalues in all 39,249 transformed matrices. At exactly zero momentum Ξ is absent, while (Q_ρ, Q_χ) reduce to the previously verified two-dimensional homogeneous physical block.

For fixed comoving momentum, $\dot q_{\rm phys} = -Hq_{\rm phys}$, so vertex projection uses $\dot y = T\dot p + \dot T p$. Cubic and quartic momentum-space kernels must be transformed leg by leg:

$$\begin{aligned} V_{abc}^{\rm phys} &= T^i{}_a(k_1)T^j{}_b(k_2)T^k{}_c(k_3)V_{ijk}, \\ W_{abcd}^{\rm phys} &= T^i{}_a(k_1)T^j{}_b(k_2)T^k{}_c(k_3)T^\ell{}_d(k_4)W_{ijk\ell}. \end{aligned} \tag{71}$$

The basis and projection map pass. Applying that map to the complete analytic interactions gives a factorized cubic kernel and a reduced quartic kernel

$$W_{\rm red} = W_{\rm contact} - \sum_{(ab|cd)} B_{ab}(-K)^{\mathsf{T}}C(K)^{-1}B_{cd}(K), \qquad (ab|cd) = (12|34), (13|24), (14|23), \tag{72}$$

where $B_{ab} = \partial^3 L_3/(\partial\epsilon_a\partial\epsilon_b\partial z_K)$ is the complete physical two-leg constraint source. The four-leg contact, the pair source, and the three Schur pairings pass in factorized form for nonzero external momenta and nonzero K.

At exact $K = 0$ one must not substitute into the finite-wavenumber inverse. The homogeneous scalar-shift coordinate $\Sigma = -D^2\beta$ is absent, just as Ξ is absent from the homogeneous physical space. For $V \neq 0$, the audited algebraic prescription is

$$P_z^{(0)} = \mathrm{diag}(1, 0), \qquad G_z^{(0)} = \mathrm{diag}\left(-\frac{1}{2V}, 0\right), \qquad P_{\mathrm{phys}}^{(0)} = \mathrm{diag}(0, 1, 1, 1), \tag{73}$$

on $(\delta N, \Sigma)$ and $(\Xi, Q_\rho, Q_\chi, \Pi)$ respectively. It retains the homogeneous lapse constraint and the (Q_ρ, Q_χ, Π) subspace while annihilating the two excluded gauge coordinates. Projector idempotence, the projected constraint-inverse identity, the three-pair combinatorics and a finite centre-of-mass lapse-source limit are verified.

The corresponding local frozen-time quadratic kernels have now been constructed. With $A_p = P_p - P_p^{\mathsf{T}}$, the finite-wavenumber convention is

$$D(\omega, q) = \omega^2 K_p + i\omega A_p + C_p, \qquad G_F = i[D + i\epsilon]^{-1}. \tag{74}$$

This includes the time derivative of the physical-basis map at fixed comoving momentum. A separate exact-$q = 0$ Hessian is formed only after the projectors in Eq. (73) are applied. Across the representative scan the kinetic inertia is positive, finite-q constraints remain nonsingular and every sampled real positive-frequency pole has positive residue. All five $q/H = 100$ snapshots have four real positive-frequency poles. Lower and intermediate momenta, however, develop complex frozen-background pole pairs, as do later projected $q = 0$ snapshots.

The required follow-on now tracks fixed comoving momentum, $q = k/a$, and evolves the complete physical equation

$$K_p\ddot{p} + \left(\dot{K}_p + 3HK_p + P_p - P_p^{\mathsf{T}}\right)\dot{p} + \left(\dot{P}_p + 3HP_p - C_p\right)p = 0. \tag{75}$$

Its kinetic-normalized transfer matrices converge with maximum coarse/fine error 1.31×10^{-4} and reproduce an independent canonical-Hamiltonian generator to 1.06×10^{-4}. Frozen-pole exponentiation is not quantitatively valid in the nonadiabatic domain. The $q/H = 100$ trajectory is a controlled high-momentum subset, with $\max|\dot{\omega}/\omega^2| = 0.0222$. The trajectory beginning at $q/H = 0.01$ nevertheless retains a converged maximum kinetic-normalized phase-space gain of 1.38×10^{27}. A five-case mode-resolved follow-on constructs kinetic-normalized pole-pair frames, assigns them by principal-angle overlap and parallel-transports their real orientations. In every tested case the maximizing initial singular vector is seeded by the finite-q Ξ gauge-continuation subspace at fraction greater than 0.99977. Each trajectory nevertheless enters an off-axis complex quartet for 3.25–3.62 percent of the sampled interval. Its real invariant space has rank four, so no unique continuous real rank-two split between the gauge-continuation and retained-matter pole pairs exists there.

The source-to-observable follow-on avoids assigning such pole identities. Generalized impulses and readouts are restricted to (Q_ρ, Q_χ); their original-field covectors annihilate the homogeneous time-translation orbit, and direct Ξ support is below 7.61×10^{-21}. All five cases retain amplified through-quartet response from 2.68×10^{17} to 9.76×10^{19}, with coarse/fine errors below 5.48×10^{-5}. Record `PASS_GAUGE_PROJECTED_MATTER_RESPONSE_SURVIVES_WITH_SCOPE`. This resolves the direct gauge-source attribution question in the tested dimensionless neighborhood, not an all-background instability theorem. The Track-A force mode Π factorizes from this coupled quadratic block and remains in the full finite-q framework.

A subsequent fixed-comoving domain map applies the declared conditions that all coupled poles remain real, $\max|\dot{\omega}/\omega^2| < 0.1$, every frequency has $|\omega|/H \geq 10$, and the factorized Π mode satisfies the same adiabatic bound. On the representative branch, samples beginning at $q/H = 47.5, 50, 75, 100$

pass, while $q/H = 45$ reaches 0.1103 and fails the adiabatic criterion. This brackets a sampled transition between 45 and 47.5; it is not a continuous or all-background boundary. The high-q frames are tracked without imposing the infrared Ξ-pure label. Each nonzero internal exchange momentum must independently pass the same domain gate, while exact $q_K = 0$ uses the separate homogeneous projector. Within those admitted samples, the analytic cubic kernel is now contracted with two residue-normalized on-shell coupled modes while its third leg remains an off-shell covector with $\dot{p}_K = -i\Omega_K p_K$. Forty-eight equilateral sum- and difference-frequency cases give finite nonzero pair sources. External-leg permutation closes to 6.11×10^{-16}, inverse response closes to 5.53×10^{-16} and the nearest sampled pole separation is 0.280. Two coupled external legs source no factorized Π component at this analytic cubic order. Record `PASS_MODE_PROJECTED_CUBIC_PAIR_SOURCE`. This is not yet an exchange amplitude: matched left/right sources, channel summation and the reduced quartic contact remain to be combined. The exact $|\nabla\pi|^3$ interaction also still requires its declared local nonzero-gradient background.

3.17 Causality and cutoff status

Around a nonzero spatial background gradient of magnitude q, the force fluctuation has

$$\omega^2 = \frac{3Aq(1+\cos^2\theta)}{K_Q}k^2 + \frac{\gamma}{K_Q M_*^2}k^4. \tag{76}$$

Thus the long-wavelength metric-light-cone threshold depends on the unmatched ratio Aq/K_Q, while phase and group velocities grow without bound at large k inside the truncated dispersion. Preferred-frame metric superluminality is not by itself a proof of acausality, but the foliation-causality test and the physical EFT domain have not been established.

Expansion of the cubic vertex supplies only the restricted, time-normalized longitudinal IR NDA scale

$$\Lambda^{(\mathrm{IR})}_{\mathrm{NDA,long}} \sim \frac{K_Q^{3/4}}{\sqrt{A}}. \tag{77}$$

It is not the physical cutoff Λ_{EFT}: transverse normalization, k^4-dominated power counting and a loop or unitarity analysis remain open. The comparison of the light-cone crossing scale with Λ_{EFT} therefore cannot yet be made.

A separate dimensional analogy, $K_Q = k_Q M_P^2$, would place the long-wavelength threshold at order a_0 for order-unity coefficients. That exercise is explicitly speculative: it is introduced only as a priority signal for deriving K_Q, not as a property of ITSM.

3.18 Conditional infrared operator

Define the normalized spatial invariant

$$Y = \frac{h^{\mu\nu}\nabla_\mu\psi\nabla_\nu\psi}{a_0^2}. \tag{78}$$

The target low-acceleration operator remains

$$\mathcal{L}_{\psi,\mathrm{IR}} = -\frac{2C_{\mathrm{IR}}}{3}M_{\mathrm{P}}^2 a_0^2 Y^{3/2} + \mathcal{O}(Y^2, \nabla^2/\Lambda_{\mathrm{EFT}}^2). \tag{79}$$

Neither C_{IR}, K_Q nor the normalized matter vertex has been derived. The preceding analytic Born–Infeld expression also remains rejected as the source of this branch because its small-invariant equation is linear at leading order. UVIR-003 is therefore in progress and MAT-001 remains blocked.

4 Conservation and syntropic throughput

The observable matter–plenum subsystem may exchange energy and momentum with a reservoir sector external to that observable subsystem but contained within the complete conserved theory. The complete system must satisfy the contracted Bianchi identity. We therefore distinguish two currents:

$$\nabla_\mu T_m^{\mu\nu} = Q^\nu_{\rm mp}, \tag{80}$$

$$\nabla_\mu T_P^{\mu\nu} = -Q^\nu_{\rm mp} + Q^\nu_{\rm syn}, \tag{81}$$

$$\nabla_\mu T_R^{\mu\nu} = -Q^\nu_{\rm syn}. \tag{82}$$

Adding Eqs. (80)–(82) gives

$$\nabla_\mu \left(T_m^{\mu\nu} + T_P^{\mu\nu} + T_R^{\mu\nu}\right) = 0. \tag{83}$$

$Q^\nu_{\rm mp}$ is the local matter–plenum exchange generated by the same interaction that sources the force scalar. $Q^\nu_{\rm syn}$ is reservoir throughput. Treating them as one undifferentiated Q^ν hides two different physical jobs and prevents a clean local/cosmological split.

Stress-energy throughput is also distinct from transfer of the condensate's $U(1)$ charge. If n is the homogeneous condensate charge density, write

$$\dot{n} + 3Hn = S_N. \tag{84}$$

No identification of S_N with $Q^\nu_{\rm syn}$ is assumed; such a relation must follow from a declared reservoir interaction.

4.1 Candidate local interaction

A universal conformal matter metric illustrates how a force and an exchange current can share an origin,

$$\widetilde{g}_{\mu\nu} = e^{2C_m\psi/c^2} g_{\mu\nu}, \qquad S_m = S_m[\widetilde{g}_{\mu\nu}, \Psi_m]. \tag{85}$$

For pressureless matter its nonrelativistic limit contains

$$\mathcal{L}^{\rm NR}_{\rm int} = -C_m \rho_b \psi, \tag{86}$$

and its exchange current is proportional to the matter trace times $\nabla^\nu\psi$. This is a candidate matter coupling, not a complete relativistic phenomenology: a purely conformal rescaling does not by itself provide the required lensing.

4.2 Reservoir throughput and anisotropic stress

In the plenum rest frame a general stress tensor includes

$$T_P^{\mu\nu} = \rho_P U^\mu U^\nu + p_P h^{\mu\nu} + \pi^{\mu\nu}. \tag{87}$$

The traceless spatial tensor $\pi^{\mu\nu}$, not the energy-transfer vector alone, carries directional stress. In a biaxial state one may define

$$\Pi = p_t - p_p, \qquad \pi^i{}_j = \mathrm{diag}\left(-\frac{2\Pi}{3}, \frac{\Pi}{3}, \frac{\Pi}{3}\right). \tag{88}$$

For the biaxial expansion used below, the shear tensor is

$$\sigma^i{}_j = \mathrm{diag}\left(-\frac{2\delta}{3}, \frac{\delta}{3}, \frac{\delta}{3}\right), \qquad \pi^{\mu\nu}\sigma_{\mu\nu} = \frac{2}{3}\Pi\delta. \tag{89}$$

The last equality is the exact anisotropic work term for these conventions. A relation such as $Q^\nu_{\rm syn} \propto \Theta U^\nu$ injects energy along the background flow but does not determine Π. A geometry-dependent action or constitutive law is required to convert throughput into shear.

No fixed redshift law for $Q^\nu_{\rm syn}$ is adopted in this alpha. Its background and perturbation dependence must follow from the reservoir model, and local decoupling must be tested rather than asserted from $\Theta \to 0$.

5 Conditional low-acceleration matter coupling

5.1 Inverse construction of the cubic-gradient operator

Nonlinear Poisson equations derived from nonquadratic potential actions have a standard precedent in AQUAL-type theories [10]. Here we isolate the coefficient normalization required by the ITSM infrared branch. Let $q = |\nabla\psi|$ and consider the static force-scalar action

$$S_\psi = \int \mathrm{d}t\,\mathrm{d}^3x\,[-K(q) - C_m\rho_b\psi]\,. \tag{90}$$

Variation yields

$$\nabla\cdot\left[\frac{K'(q)}{q}\nabla\psi\right] = C_m\rho_b. \tag{91}$$

The required cubic-gradient operator is parameterized as

$$K(q) = \frac{C_{\rm IR}}{12\pi G a_0}q^3, \qquad C_{\rm IR} > 0. \tag{92}$$

The complete static weak-field action in this normalization is

$$\begin{aligned} S_{\rm WF} = \int \mathrm{d}t\,\mathrm{d}^3x \Bigg[&-\frac{|\nabla\Phi_N|^2}{8\pi G} - \rho_b\Phi_N \\ &- \frac{C_{\rm IR}}{12\pi G a_0}|\nabla\psi|^3 - C_m\rho_b\psi\Bigg]. \end{aligned} \tag{93}$$

For independent positive coefficients C_m and $C_{\rm IR}$, the sourced equation is

$$\nabla\cdot\left[\frac{|\nabla\psi|}{a_0}\nabla\psi\right] = 4\pi G\frac{C_m}{C_{\rm IR}}\rho_b. \tag{94}$$

For spherical symmetry this gives

$$|\nabla\psi|^2 = \frac{C_m}{C_{\rm IR}}a_0 g_N. \tag{95}$$

Since the physical acceleration is $g_P = C_m|\nabla\psi|$, the invariant observable square-root coefficient is

$$g_P = C_{\rm obs}\sqrt{a_0 g_N}, \qquad \boxed{C_{\rm obs} = \frac{C_m^{3/2}}{\sqrt{C_{\rm IR}}}}. \tag{96}$$

Only under the normalization choice $C_m = C_{\rm IR} = C$ does $C_{\rm obs} = C$. That convenient choice is not a physical derivation of $C = 2/3$; MAT-001 must calculate the invariant combination in Eq. (96).

5.2 Compact toroidal source

On compact T^3 there is no boundary at spatial infinity, and a periodic divergence integrates to zero. Write

$$\rho_b = \bar{\rho}_b + \delta\rho_b, \qquad \psi = \bar{\psi}(t) + \varphi(t, \mathbf{x}), \tag{97}$$

with zero spatial averages for $\delta\rho_b$ and φ. The local periodic equation has the form

$$D_i \left[\frac{|D\varphi|}{a_0} D^i \varphi \right] = 4\pi G \frac{C_m}{C_{\rm IR}} \delta\rho_b + \mathcal{S}_Q, \qquad \int_{T^3} \left(4\pi G \frac{C_m}{C_{\rm IR}} \delta\rho_b + \mathcal{S}_Q \right) \mathrm{d}V = 0. \tag{98}$$

Here $\mathcal{S}_Q \equiv \mathcal{S}_Q[\delta Q_{\rm syn}]$ is a conditional zero-mean scalar source representing the inhomogeneous projection of reservoir–plenum exchange into the force-scalar equation. Its functional map from the vector exchange current has not been derived; setting $\mathcal{S}_Q = 0$ is the local-static baseline rather than a claim that reservoir perturbations vanish identically. The homogeneous mode belongs to the cosmological/open-system problem. A localized spherical solution is therefore the small-region, compensated-overdensity limit of Eq. (98), not an isolated global cosmology.

5.3 Disk geometry and coefficient status

For a general density distribution, equality of divergences permits

$$\frac{|\nabla\varphi|}{a_0} \nabla\varphi = \frac{C_m}{C_{\rm IR}} \nabla\Phi_N + \nabla \times \mathbf{H}. \tag{99}$$

The solenoidal field vanishes in sufficient symmetry but is generally nonzero for a finite disk. Consequently a pointwise square-root prescription is a historical algebraic approximation. DISK-001 must solve Eq. (98) for realistic baryonic sources before $C_{\rm obs} = 2/3$ and $C_{\rm obs} \simeq 1$ can be fairly compared.

The separate coefficients change under a rescaling of ψ, while the physical response is carried by the invariant combination $C_{\rm obs} = C_m^{3/2}/\sqrt{C_{\rm IR}}$. The trace ratio 2/3 remains a geometric motivation and a matching hypothesis, not a completed local Wilson coefficient.

6 Toroidal topology and validated Casimir backreaction

6.1 Periodic rectangular lattice

For a massless scalar with periodic lengths (L_1, L_2, L_3), the renormalized energy density used in CBR-001 is the rectangular-periodic Epstein-lattice result developed in Ref. [11]:

$$\rho_{\rm Cas} = -\frac{\hbar c}{2\pi^2} \sum_{\mathbf{n}\in\mathbb{Z}^3\setminus\{0\}} \frac{1}{[(n_1L_1)^2 + (n_2L_2)^2 + (n_3L_3)^2]^2}. \tag{100}$$

The directional pressures follow thermodynamically,

$$p_i = -\frac{1}{L_jL_k} \frac{\partial E_{\rm Cas}}{\partial L_i}, \qquad E_{\rm Cas} = L_1L_2L_3\rho_{\rm Cas}. \tag{101}$$

The Stage-1 and Stage-2 calculations validate cutoff convergence, cubic isotropy, pressure derivatives, the massless trace relation, dimensional scaling and the sign change of $p_t - p_p$ across the cubic shape. Thus compact shape-dependent anisotropic stress is a concrete mechanism. It is not a universal cycle-counting coefficient. A recent direct semiclassical calculation likewise derives a finite topology-dependent stress tensor on T^3 and its anisotropic de Sitter backreaction [14].

6.2 Biaxial backreaction

For

$$ds^2 = -dt^2 + a_p^2 dx^2 + a_t^2(dy^2 + dz^2), \tag{102}$$

define

$$H = \frac{H_p + 2H_t}{3}, \quad \delta = H_t - H_p, \quad r = \frac{a_t}{a_p}. \tag{103}$$

Then

$$\dot{\delta} + 3H\delta = \kappa(p_t - p_p), \qquad \dot{r} = r\delta, \tag{104}$$

with the Hamiltonian constraint

$$3H^2 - \frac{\delta^2}{3} = \kappa\rho_{\text{total}}. \tag{105}$$

These are the biaxial specialization of the standard Bianchi-I system; see, for example, Ref. [12]. The Stage-3A solver reproduces the perturbative solution and preserves the constraint and Casimir continuity equation to substantially better than the specified tolerances.

6.3 The Stage-3B ratio test

Let

$$x = \frac{\delta}{H}, \qquad q = \frac{H_t}{H_p} = \frac{1 + x/3}{1 - 2x/3}. \tag{106}$$

The historical target $q = 13/12$ corresponds to

$$x_\star = \frac{3}{38}. \tag{107}$$

The small-source fixed-background solution is

$$x(N) = S\left(e^{-3N} - e^{-4N}\right), \tag{108}$$

with maximum $(27/256)S$ at $N = \ln(4/3)$. Merely reaching $x_\star$ therefore requires

$$S \geq \frac{128}{171} \simeq 0.748538, \tag{109}$$

already outside the strict small-perturbation regime.

The nonlinear root search finds target-reaching amplitudes for five of the tested initial shapes. Every case is classified as a transient crossing, with only approximately 0.27–0.30 e-folds within one percent of the target. Four thresholds are nonperturbative and one is marginal by the stated Casimir-fraction diagnostic. Every valid candidate returns to $q = 1$ by $N = 20$; no quasi-plateau or attractor is found.

There is also a kinematic warning. Since

$$\frac{d\ln r}{dN} = x, \tag{110}$$

a constant $q \neq 1$ implies continuous evolution of the shape ratio. It cannot be a fixed torus shape. A persistent anisotropic state would have to be a driven solution, not a static geometric ratio.

For a desired trajectory $x(N)$, the required anisotropic stress is

$$\Pi_{\text{req}} = \frac{H^2}{\kappa}\left[\frac{dx}{dN} + x\left(3 + \frac{d\ln H}{dN}\right)\right]. \tag{111}$$

In de Sitter with constant $x = 3/38$ this becomes

$$\Pi_{\text{req}} = \frac{9H^2}{38\kappa}. \tag{112}$$

For the CBR-001 physical lengths $L_p = L_\star a r^{-2/3}$ and $L_t = L_\star a r^{1/3}$, the anisotropic Casimir source is more precisely

$$\Pi_{\text{Cas}}(a, r) = \frac{\hbar c}{L_\star^4} a^{-4} r^{8/3} \left[\widehat{p}_t(r) - \widehat{p}_p(r)\right] \equiv \frac{\hbar c}{L_\star^4} a^{-4} F_\Pi(r). \tag{113}$$

Thus the explicit scale dependence is radiation-like, while the shape function evolves whenever r evolves. In every validated free-field trajectory the source nevertheless becomes negligible and cannot maintain Eq. (111). A future CBR-002 test must derive any additional shape, driven or memory stress without inserting the target ratio into a coefficient.

7 Wake, circulation, and non-equilibrium hypotheses

The original ITSM described baryonic matter as exciting a torsional or acoustic wake in a rotational medium. The static equation Eq. (98) has no independent memory: for specified boundary data it tracks the instantaneous source. A persistent, advected or detached wake therefore requires additional dynamics.

A schematic relaxation equation is

$$\tau_W U^\mu \nabla_\mu W + W = \mathcal{F}[\psi, \rho_b, r, Q_{\text{syn}}], \tag{114}$$

or, if wave propagation is essential, a hyperbolic field equation with a second time derivative. Equation (114) is not adopted as a law. It identifies the quantities that WAK-001 must derive or bound. Its first-order form is motivated only as a causal-relaxation template of the Israel–Stewart type [13]; it is not a derivation of an ITSM constitutive law.

A viable wake sector must satisfy all of the following:

1. a well-posed initial-value problem;
2. causal characteristic speeds in the physical frame;
3. a positive or bounded energy accounting;
4. a controlled static limit reproducing the IR field equation;
5. explicit momentum exchange with matter and the reservoir;
6. decay or transport laws rather than indefinite unsupported memory.

Similarly, circulation is represented naturally by the phase in Eq. (2),

$$\oint \nabla_i \Theta \, \mathrm{d}x^i = 2\pi n, \qquad n \in \mathbb{Z}, \tag{115}$$

where zeros of ρ permit defect cores. The relationship between these microscopic winding sectors and a global toroidal current is the VOR-001 gate. No black-hole, lensing or galactic coefficient follows from the winding number without further dynamics.

For driven anisotropy one may organize the bookkeeping as

$$\Pi_{\text{total}} = \Pi_{\text{Cas}} + \Pi_{\text{shape}} + \Pi_{\text{driven}} + \Pi_{\text{memory}}. \tag{116}$$

Only Π_{Cas} has been calculated in the present programme. The other terms are labels for research sectors. Writing Eq. (116) does not establish that they exist or have the required sign and magnitude.

8 Cosmological architecture and current limits

The homogeneous condensate, reservoir and topology moduli must define one fiducial cosmological background before CMB, growth, age or distance results are calculated. Earlier analyses used several dataset-conditioned values of the syntropic exponent or effective equation of state and sometimes mixed them across diagnostics. Those outputs remain provenance and cannot be treated as one canonical cosmology.

At background level, a conservative sector decomposition has the form

$$\dot{\rho}_P + 3H(\rho_P + p_P) + \pi^{\mu\nu}\sigma_{\mu\nu} = Q, \tag{117}$$

$$\dot{\rho}_R + 3H(\rho_R + p_R) = -Q, \tag{118}$$

For the biaxial conventions of Sec. 6, Eq. (89) gives the exact specialization

$$\dot{\rho}_P + 3H(\rho_P + p_P) + \frac{2}{3}\Pi\delta = Q. \tag{119}$$

The total continuity equation follows by summing the sectors.

The UVIR-003 background-completion screen selects this evolving cosmological class over an artificially supported exact-Minkowski finite-density state. A representative dimensionless on-shell branch now establishes existence, but does not provide a fiducial parameter point or cosmological fit. In particular, isolated condensate charge dilutes according to Eq. (29); holding it constant would require the distinct source in Eq. (84).

No predetermined law such as $Q \propto (1+z)^{-3}$ is adopted here. In particular, $(1+z)^{-3} = a^3$ grows toward late time and cannot simultaneously be described as an early-universe enhancement. COS-001 must derive the source and fit one background with an explicit parameter inventory.

8.1 What CBR-001 changes

CBR-001 does not remove topology-dependent cosmological stress. It shows that the tested passive free-field source has the evolving form $\Pi_{\mathrm{Cas}} = a^{-4}F_{\Pi}(r)$ and becomes unable to support a persistent 13/12 directional expansion ratio. Therefore

- $H_0 = 72.97\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is not a parameter-free ITSM prediction;
- assigning Planck and SH0ES to opposite directional extrema is not a derived observation model;
- any future directional expansion must be passed through actual survey window functions;
- a driven anisotropic state requires a derived source beyond free Casimir stress.

8.2 Perturbations

PERT-001 must vary the same total action and exchange currents used for the background. Effective CAMB or CLASS mappings are useful diagnostics, but they do not replace the derivation of scalar, vector and tensor perturbations, initial conditions, stability and observable transfer functions. Until that gate closes, CMB-peak and S_8 values are diagnostic calculations rather than canonical predictions.

9 Dependency-ordered closure gates

Table 2: Research gates required before restoration of strong predictions.

Gate and status	Question	Pass condition
UIVR-001 *Closed negative*	Does the minimal condensate yield spatial $Y^{3/2}$?	Candidate rejected; replacement route required
UVIR-002 *Closed provisional*	Which replacement architecture survives initial checks?	Two-sector preferred-frame route selected for testing
UVIR-003 *In progress*	Is the selected action stable, causal and weakly coupled?	Full coupled constraint reduction, physical cutoff and background tests
MAT-001 *Blocked*	What is the normalized matter vertex?	Force and exchange from one action; C_m, C_{IR} and K_Q matched
SCR-001 *Open*	Is the force screened at high gradient?	Ephemeris and binary constraints satisfied
LEN-001 *Open*	What metric do matter and light see?	PPN, lensing and wave propagation derived
DISK-001 *Open*	What is the exact disk response?	Validated periodic nonlinear solver and curl field
TOP-001 *Open*	How do shape moduli evolve?	Covariant moduli action and stable dynamics
VOR-001 *Open*	How is global circulation carried?	Winding and defect sector with finite energy
WAK-001 *Open*	Can wakes retain causal memory?	Hyperbolic or relaxation model with energy accounting
CBR-002 *Open*	Can derived sectors sustain anisotropy?	Untuned non-equilibrium solution and stability test
COS-001 *Open*	What is the fiducial background?	One coherent fit from declared equations
PERT-001 *Open*	What are the perturbation equations?	Action-derived stable transfer system

The dependency order is

$$\begin{aligned}
\text{UVIR-001} \to \text{UVIR-002} \to \text{UVIR-003} \to \text{MAT-001}& \\
\to (\text{SCR}, \text{LEN}) \to \text{DISK}& \\
\to (\text{TOP}, \text{VOR}, \text{WAK}, R) \to \text{CBR-002}& \\
\to (\text{COS}, \text{PERT}).& \qquad (120)
\end{aligned}$$

The frame-sector literature substitution, zero-gradient force-block factorization, ADM readiness audit and longitudinal IR NDA estimate are progress within UVIR-003, not new closed gates. The force block contains one nonnegative $z = 2$ scalar in the declared zero-gradient truncation, while a rescaling audit shows that standalone K_Q is not identifiable before physical field normalization. The declared finite-density Minkowski background is off shell unless a non-vacuum support sector is supplied; its scalar response must be included before the remaining lapse and shift constraints are eliminated. The minimal support screen rejects vacuum energy, a healthy two-derivative $P(X)$ scalar and the ghost-condensate point as exact Minkowski completions; rigid support is decoupling-only. An evolving flat-FRW minisuperspace system and a representative on-shell branch are now verified. A first aether-unitary ADM subgate eliminated the principal lapse and scalar-shift constraints and independently recovered the exact spin-0 aether speed. The complete quadratic finite-q reduction has now also eliminated the constraints along that branch. Its 48,861 sampled nonzero-wavenumber kinetic matrices have three positive and no negative eigenvalues. The exact determinant is proportional to q^2_{phys}, but the follow-on endpoint audit proves that the collapsing direction is $(H, \dot{\rho}, \mu)$, the homogeneous time-translation orbit. The two independent gauge-invariant matter scalars retain a positive regular $q = 0$ kinetic block across the representative trajectory. Thus the rank loss is a gauge endpoint, not a physical ghost or a third homogeneous scalar. The full three-dimensional flat-decoupling khronon cubic and quartic operator bases are now derived. The quartic expansion contains 96 monomials and reduces to an independent longitudinal calculation. Its elastic contact term is finite, while elastic t/u cubic exchange vanishes exactly. The centre-of-mass s channel carries zero spatial momentum and is the non-invertible homogeneous khronon gauge orbit. Quartic reduction also requires the second-order constraint Schur complement, although third-order constraint solutions cancel. The exact nonlinear $g + U + \Phi$+alignment ADM parent block reproduces the FRW and finite-q quadratic constraint system. Track A now adopts the rest-space Laplacian and retains exact $Y^{3/2}$ for a declared local nonzero-gradient force analysis. Its homogeneous force action is verified through direct quartic order. Its origin-linear J_2 is a verified component, while the complete dressed source is $S_2 = \partial_z L_3[x, z_1]$. The generic gravity–aether and condensate dressing, complete S_2, generic $L_4[x, z_1]$ contact and corrected reduced quartic functional are assembled at $q_{\mathrm{phys}} > 0$. A regular basis (Ξ, Q_ρ, Q_χ) has a finite positive low-wavenumber kinetic limit over the validated representative domain and excludes the exactly homogeneous Ξ gauge mode. The time-dependent leg-wise projection map is fixed. The complete analytic cubic and reduced quartic functionals are now polarized into factorized finite-q physical-basis kernels. The quartic construction includes the complete two-leg source and all three constraint Schur pairings. At exact zero internal momentum, audited projectors remove $\Sigma = -D^2\beta$ and the homogeneous Ξ time-translation orbit while retaining the lapse constraint and (Q_ρ, Q_χ, Π). This rule is not a naive $q \to 0$ substitution. The finite-q physical inverse quadratic kernel and separate exact-$q = 0$ projected response are now constructed. Positive kinetic inertia, nonsingular constraints, inverse closure and positive residues for sampled real positive-frequency poles are verified. The $q/H = 100$ subset is a clean four-mode real-pole regime. The required fixed-comoving follow-on restores coefficient derivatives and Hubble dilution and yields converged kinetic-normalized transfer matrices. Frozen-pole exponentiation is invalid in the nonadiabatic domain. The trajectory beginning at $q/H = 0.01$ still has a maximum normalized phase-space gain of 1.38×10^{27}. Five nearby mode-resolved cases all have an Ξ-seeded maximizing initial vector, but all enter an off-axis complex quartet whose real invariant space is rank four. The nominal pole pairs therefore have no unique continuous real rank-two split through that interval. The follow-on retarded-response audit instead applies generalized impulses and readouts only in retained (Q_ρ, Q_χ), with no direct Ξ or homogeneous time-translation source support. All five cases retain amplified through-quartet response from 2.68×10^{17} to 9.76×10^{19}. This passes the bounded gauge-source attribution subcalculation but is not an all-background instability theorem. The factorized Track-A force mode remains in the full finite-q framework outside the coupled quartet block. A controlled sampled exchange domain now exists on the representative branch:

initial $q/H = 47.5, 50, 75, 100$ passes the real-pole, subhorizon and 0.1 adiabatic criteria, whereas $q/H = 45$ fails the adiabatic threshold. Every nonzero internal channel must pass independently, and exact $q_K = 0$ retains its separate projector. Across 48 admitted equilateral mode/sign cases, two residue-normalized on-shell legs now produce finite nonzero cubic pair-source covectors and finite inverse-kernel responses, with permutation and inverse closure below 6.2×10^{-16}. Matched left/right pair-source contraction, channel summation and the reduced quartic contact are not yet combined. The exchange-plus-contact amplitude, unitarity scale, and physical cutoff are still open. The representative spin-0 cone is wider than the metric cone. The physical EFT cutoff, global multicone causality and nonzero-gradient force mixing remain unresolved. Likewise, running CBR-002 with a source adjusted to hold 13/12 would test a tuning procedure, not ITSM, and fitting the algebraic disk law does not answer the periodic nonlinear field problem.

10 Falsifiability without overclaiming

The recovery replaces spectacular but weakly derived forecasts with a sequence of gate-level falsifiers.

10.1 Near-term theoretical falsifiers

1. UVIR-001 has already rejected the minimally kinetic condensate as the direct origin of the spatial cubic law. If UVIR-003 cannot produce a stable, causal and weakly coupled two-sector completion with a controlled physical cutoff, the selected replacement architecture is rejected as well.

2. If no matching calculation fixes K_Q, C_{IR} and the normalized matter vertex in a domain compatible with the mode constraints, MAT-001 cannot open and the local force law remains an uncompleted EFT ansatz.

3. If a normalized matter vertex cannot satisfy galactic and local bounds within one action, MAT-001/SCR-001 reject the universal coupling architecture.

4. If no relativistic completion produces acceptable PPN and lensing potentials, the local acceleration law cannot serve as a gravitational model.

5. If the periodic disk solution preserves the empirical tension of the algebraic $C_{\mathrm{obs}} = 2/3$ implementation, disk geometry cannot rescue that matching hypothesis.

6. If derived reservoir, shape and wake sectors have no stable driven anisotropic solution, CBR-002 rejects persistent topological Hubble anisotropy within that architecture.

10.2 Observational claims held downstream

NANOGrav spectral features require a derived eigenmode spectrum and detector response. Cluster offsets require causal wake dynamics, a collision calculation and a lensing metric. Stellar-population claims require an independent star-formation model. CMB and growth require a coherent background and perturbation theory. None is presented here as a clean framework-level falsifier.

10.3 Reproducibility rule

Every future numerical claim must identify its source equations, code entry point, data version, parameter inventory, validation controls and a machine-readable summary. A reported statistic must be explicitly computed by the cited pipeline. Negative and partial gate results remain part of the model record rather than being hidden by a new ansatz.

Conclusion

The recovery preserves the defining ITSM idea—a driven, finite-density, rotational medium with compact topology and sector exchange—while recording where candidate mechanisms fail. The minimal condensate route to the spatial cubic law is closed negative; the replacement preferred-frame force sector has passed bounded Stage-A, frame-subsystem, scalar principal-symbol and finite-q constraint checks, and its homogeneous rank-loss direction is now identified as a time-translation gauge orbit. The complete three-dimensional flat-decoupling khronon cubic and quartic bases and their constraint-order identities are also derived. The quartic contact and elastic t/u identities pass, and the complete cosmological cubic and reduced-quartic interactions now have factorized physical-basis momentum kernels. Separate exact-zero-momentum projectors remove the homogeneous scalar shift and Ξ gauge coordinates before the centre-of-mass channel is assembled. This closes the kernel and algebraic prescription steps. The physical inverse quadratic kernels are constructed, and the converged fixed-comoving transfer shows that frozen-pole exponentiation fails in the nonadiabatic domain. The deepest infrared trajectory retains large normalized phase-space gain. A five-case follow-on finds that its maximizing initial vector is Ξ seeded throughout the tested neighborhood, but each trajectory enters an off-axis complex quartet. The rank-four real invariant space prevents a unique continuous rank-two pole split there. A source-to-observable follow-on then restricts generalized impulses and readouts to retained (Q_ρ, Q_χ), with no direct Ξ or homogeneous time-translation source support. All five tested cases retain amplified through-quartet matter response; the baseline value is 1.43×10^{19}. This passes the bounded gauge-source attribution subcalculation without establishing an all-background instability theorem. The factorized Track-A force mode remains in the full finite-q framework outside this coupled quadratic audit. Nonlinear ADM provenance passes for the $g + U + \Phi$+alignment block, and Track A supplies the complete dressed source entering the quartic Schur term. A controlled real-pole exchange domain, the gauge-regular exchange-plus-reduced-contact scalar $2 \to 2$ amplitude, global multicone causality, matching and the physical cutoff remain open. The conditional $Y^{3/2}$ weak-field branch and the rectangular-T^3 Casimir tensor are useful controlled results, but future predictive claims must remain downstream of the unresolved gates.

References

[1] D. T. Son, "Low-energy quantum effective action for relativistic superfluids," arXiv:hep-ph/0204199 (2002), https://arxiv.org/abs/hep-ph/0204199.

[2] D. T. Son and M. Wingate, "General coordinate invariance and conformal invariance in nonrelativistic physics: Unitary Fermi gas," *Annals of Physics* **321**, 197–224 (2006), doi:10.1016/j.aop.2005.11.001, https://arxiv.org/abs/cond-mat/0509786.

[3] C. Eling, T. Jacobson and D. Mattingly, "Einstein–Aether Theory," arXiv:gr-qc/0410001 (2004), https://arxiv.org/abs/gr-qc/0410001.

[4] T. Jacobson, "Extended Hořava gravity and Einstein–aether theory," arXiv:1001.4823 (2010), https://arxiv.org/abs/1001.4823.

[5] D. Blas, O. Pujolàs and S. Sibiryakov, "Models of non-relativistic quantum gravity: The good, the bad and the healthy," arXiv:1007.3503 (2010), https://arxiv.org/abs/1007.3503.

[6] A. Papazoglou and T. P. Sotiriou, "Strong coupling in extended Hořava–Lifshitz gravity," arXiv:0911.1299 (2009), https://arxiv.org/abs/0911.1299.

[7] A. Vikman, "Can dark energy evolve to the Phantom?," *Physical Review D* **71**, 023515 (2005), doi:10.1103/PhysRevD.71.023515, https://arxiv.org/abs/astro-ph/0407107.

[8] N. Arkani-Hamed, H.-C. Cheng, M. A. Luty and S. Mukohyama, "Ghost Condensation and a Consistent Infrared Modification of Gravity," *Journal of High Energy Physics* **05**, 074 (2004), doi:10.1088/1126-6708/2004/05/074, https://arxiv.org/abs/hep-th/0312099.

[9] P. Creminelli, M. A. Luty, A. Nicolis and L. Senatore, "Starting the Universe: Stable Violation of the Null Energy Condition and Non-standard Cosmologies," *Journal of High Energy Physics* **12**, 080 (2006), doi:10.1088/1126-6708/2006/12/080, https://arxiv.org/abs/hep-th/0606090.

[10] J. Bekenstein and M. Milgrom, "Does the missing mass problem signal the breakdown of Newtonian gravity?," *Astrophysical Journal* **286**, 7–14 (1984), doi:10.1086/162570.

[11] A. Edery, "Multidimensional cut-off technique, odd-dimensional Epstein zeta functions and Casimir energy of massless scalar fields," *Journal of Physics A* **39**, 685–712 (2006), doi:10.1088/0305-4470/39/3/017, https://arxiv.org/abs/math-ph/0510056.

[12] G. F. R. Ellis and M. A. H. MacCallum, "A class of homogeneous cosmological models," *Communications in Mathematical Physics* **12**, 108–141 (1969), doi:10.1007/BF01645908.

[13] W. Israel and J. M. Stewart, "Transient relativistic thermodynamics and kinetic theory," *Annals of Physics* **118**, 341–372 (1979), doi:10.1016/0003-4916(79)90130-1.

[14] A. Negro *et al.* (COMPACT Collaboration), "Quantum signatures of cosmic topology: How Casimir backreaction transmits isotropy violation," arXiv:2603.12319 (2026), https://arxiv.org/abs/2603.12319.